

INTRODUCTION

- The majority of weather forecasts are created using Numerical Weather Prediction (NWP) models, which assimilate data from multiple sources and compute various physics-based equations to attempt to model the fluid dynamics of the atmosphere and oceans from the current state of the atmosphere to one timestep in the future.
- The conditions from that previously-computed timestep are then inputted to recursively derive the desired forecast. Small inaccuracies of each forecast multiply over time leading to increasing error as the number of timesteps in the future increases. Many of the processes that contribute to precipitation cannot be directly computed using physics equations and must be estimated, therefore making predicting precipitation and precipitation type a challenge even with advancements in NWP.
- Deep learning (DL) algorithms train input and output nodes in multiple layers through statistical techniques to identify patterns in data, leading to predictive analysis. These layers form a neural network which can represent non-linear relationships and handle high-dimensional data from multiple sources. Convolutional neural networks (CNNs) are a form of neural network that handles spatial data by including local patterns and identifying features and relationships within the spatial structure of input data.
- This research assesses the accuracy of CNN methods to predict precipitation by creating a dual CNN model using a U-Net architecture, which has historically been employed for image segmentation (Ronneberger et al., 2015). The dual U-net CNN model follows the architecture laid out by Badrinath et al. 2023, and is composed of one binary classification model used to predict the probability of precipitation at each grid point and one regression model which outputs a numerical value for total precipitation.

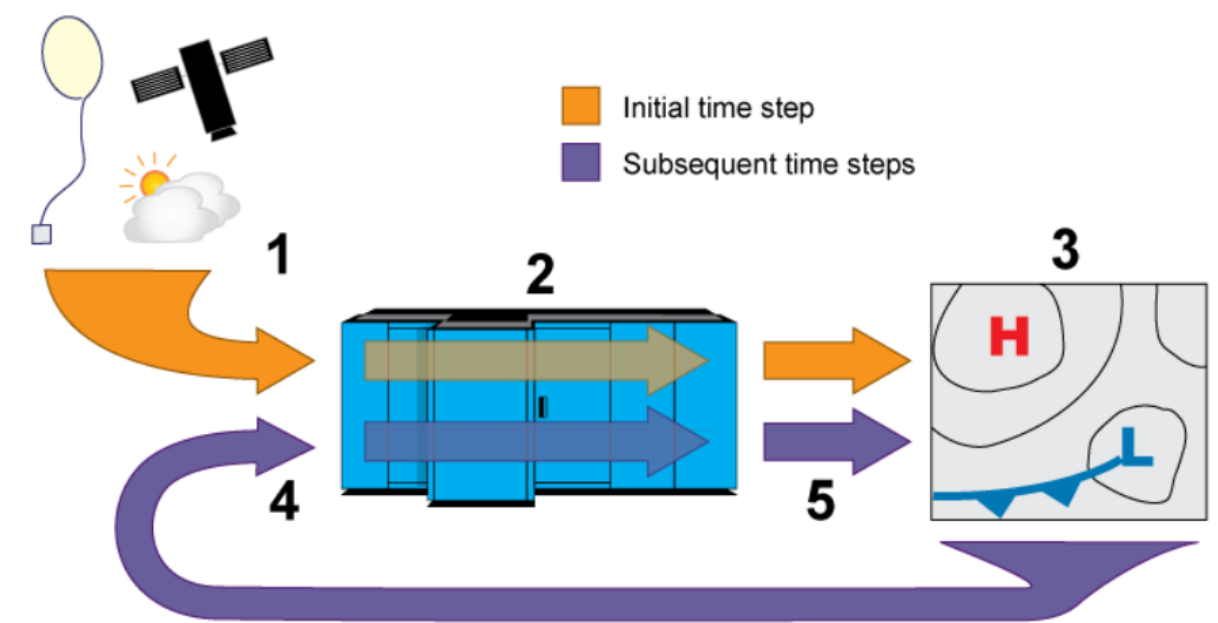


Figure 1. Simple Representation of NWP processing (NOAA 2023).

MATERIALS

- The ERA5 (hourly data on single levels) reanalysis dataset, which attempts to align weather forecasts closer to observational data, is one of the highest-resolution datasets and was chosen as input to our U-Net CNN.
- The ECMWF CDS API was used to download data at a 0.25-degree resolution for all hours of all days of December, January, February, and March for years 2000-2018 for latitudes 44 to 41 N and longitudes -75 to -72 W. This domain encompasses the majority of Central and East New York State and parts of neighboring states.
- The Weather Research and Forecasting Advanced Research (WRF-ARW) model was chosen for comparison with the U-net CNN output for a specific case study in forecasting precipitation type.
- The model was created using Google Collab and Keras Tensorflow API.

METHODS

- A pairwise Pearson correlation was computed using the Pandas library to select variables for input to predict precipitation. Variables with the highest correlation to total precipitation and their correlation coefficients can be found in the table below.
- Unexpectedly, certain variables such as dew point temperature and 2m temperature, which are used frequently in the case of precipitation prediction via U-net CNN, did not have a high correlation coefficient (Badrinath et. al, 2023). These variables were also selected for input due to their relevance in the literature, along with others such as snow depth, which was chosen for its relation to the winter months that compose the dataset.
- There exists a major class imbalance of total precipitation values as the clear majority of all values is 0, meaning no precipitation. Shown below as a histogram of precipitation values in Figure 2.

Variable	Correlation Coefficient (-1 to 1)
Total Column Water Vapor	0.445041
Surface Thermal Radiation	0.413802
Top Thermal Radiation	0.401470
Downward Thermal Radiation	0.397022
10m Wind Gust	0.350296
Surface Thermal Radiation Clear Sky	0.322620
Convective Available Potential Energy	0.312174

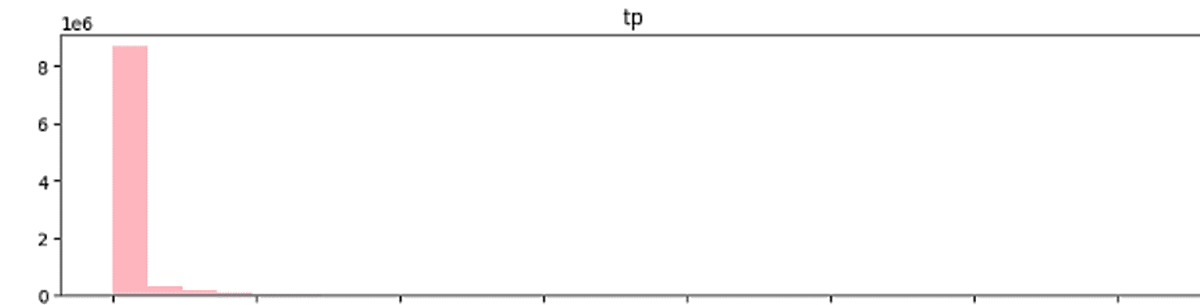
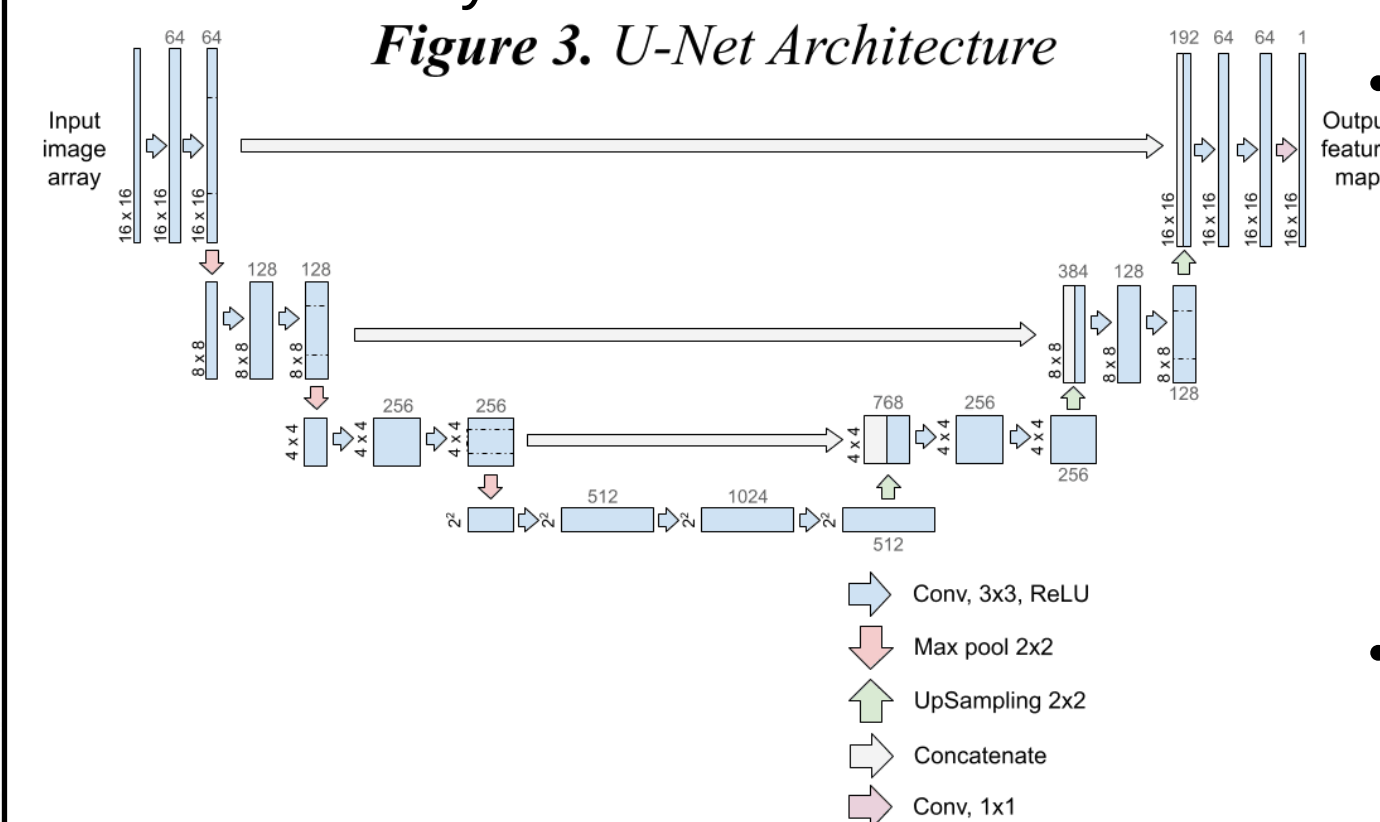


Figure 2. Histogram of Total Precipitation Values in the Domain

- To deal the imbalance of total precipitation (tp) values, the dual-binary-classification-regression model was employed, proposed by Badrinath et al. 2023. The binary classification model's predictions are then thresholded at a value of 0.5 and converted into a binary mask that is multiplied by the final output of the regression model in our combined model, effectively masking out values from the regression model where the classification model does not predict a high likelihood of rain.
- The total dataset was split into a training, validation, and test dataset based on years leading from past to present. The training dataset comprised ~80% of all hours. The validation dataset comprised ~15% of the remaining hours and months relating to a major NWP forecasting failure made up the test dataset.
- The data was transposed to shape (hours, width(longitude), height(latitude) channels) so that each variable was assigned a separate "channel" dimension for input to the U-net model.
- The numerical channels were normalized using Min-Max normalization, a common normalization technique that follows the formula below.

$$x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

- When selecting features and target variables, features encompass arrays with all 13 selected variables, while targets are defined as arrays, at an index 12 hours into the future, with only total precipitation as the variable/channel. Targets were then converted to binary: 1 where tp was greater than 0, and otherwise 0, for input in the binary model.



Binary: Activation: Sigmoid Loss: Binary Cross-Entropy (BCE)

Regression: Activation: Rectified Linear Unit (ReLU) Loss: Mean Squared Error (MSE)

This research utilizes pre-built layers through Keras including: input, 2D convolutional, max pooling, up sampling, and Lambda (used for creating a binary mask and element-wise matrix multiplication).

- The U-Net architecture and filters of each layer can be seen in Figure 3, below. The binary and regression models have the same architecture and only differ in activation and loss functions:

RESULTS

- A dataset from January 1, 2018, to March 31, 2018, was used to evaluate the performance of the model.
- A Receiver Operating Characteristic (ROC) curve is a graph used to evaluate the performance of a binary classification model by plotting the true positive rate against the false positive rate at different classification thresholds. The figure below shows the ROC curve for the binary classification model when predicting the test dataset.

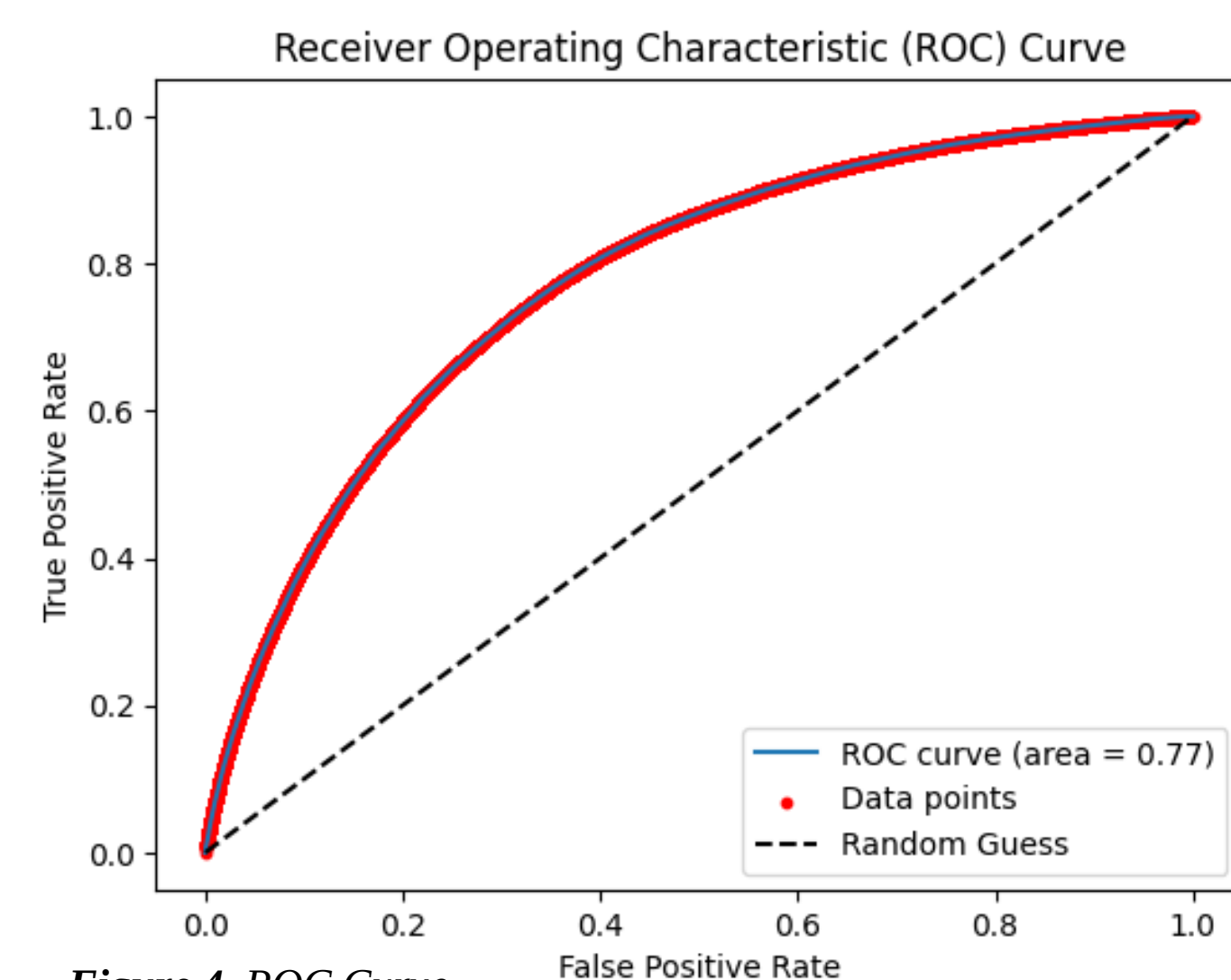


Figure 4. ROC Curve

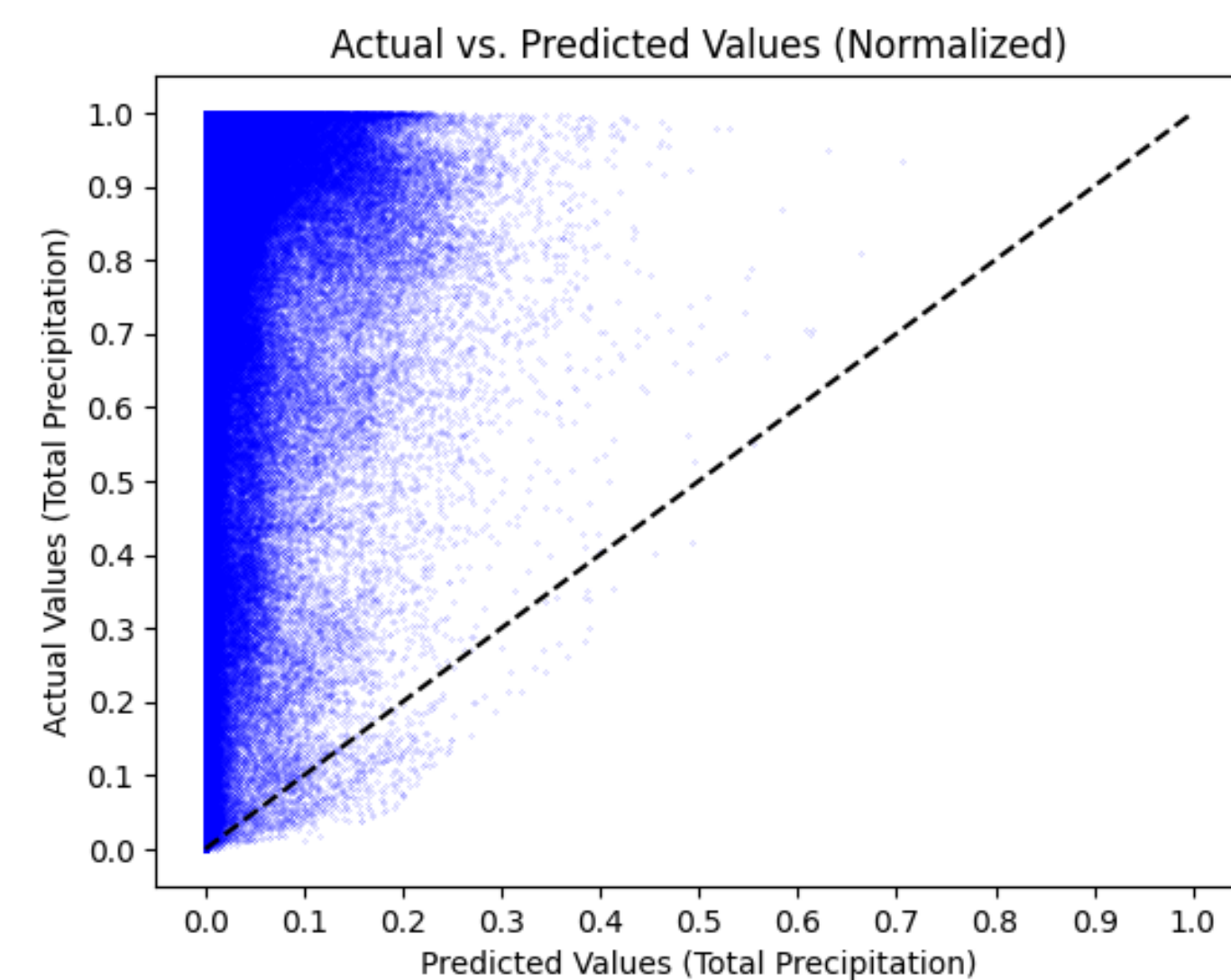


Figure 5. Actual vs. Predicted Scatterplot

- A ROC Curve positioned closer to the top-left corner of the graph, indicates a high area under the curve value, in this case, 0.77, this suggests that the binary classification portion of the combined model is performing correctly.

- Using ERA5 reanalysis as y-actual and the model predictions as y-predicted. The scatter plot is shown below. A perfect fit would be a diagonal line from the bottom left corner to the top right corner of the plot.

- The numerical half of the dual model in both its loss values per epoch and in the scatterplot, left, shows little accuracy.

- Most predicted values are 0, only few extreme actual values relate in predicted values greater than 0.1.

- Future work will focus on eliminating the gross underprediction seen and dealing with the class imbalance.

FUTURE WORK

Future work will involve:

- Using a grid of interpolated data variables from weather station networks in New York and New Jersey created through an interpolation model on ArcGIS Pro as the ground truth to post-process NWP forecasts and reanalysis.
- Using Multi-Class classification to identify precipitation type and predict precipitation type at that grid point with a separate model trained on all precipitation types
- Investigating other methods to deal with class imbalance
- Including a wider input window, multiple hourly arrays to predict precipitation at one hour some forecast hours in the future, instead of just one time (hour) as the model currently employs.
- Seeing if other CNN model architectures such as Res-Net or LSTM could be useful for taking into the temporal trends of precipitation.

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